

Quantum Tunneling of Wave Packets in One Dimension

Bryan Tiamson

Department of Physics and Astronomy, University of Delaware, Newark, DE 19716-2570, USA

This paper investigates the phenomenon of quantum tunneling through a potential barrier in one-dimensional systems by solving the time-dependent Schrödinger equation. We focus on the conditions under which a wave packet, with initial kinetic energy lower than the potential barrier, can still transmit through the barrier.

I. INTRODUCTION

Quantum tunneling demonstrates particles' ability to penetrate potential barriers with kinetic energy lower than the barrier height, a phenomenon absent in classical physics. This project focuses on tunneling in a one-dimensional system by solving the time-dependent Schrödinger equation. The study explores how a wave packet with insufficient classical kinetic energy can transmit through a potential barrier.

This research not only deepens the understanding of quantum tunneling but also examines its applications in fields like semiconductor physics and quantum computing, where tunneling is essential. By modeling the interaction of a Gaussian wave packet with a defined potential barrier, this project aims to provide insights into the probabilistic nature of quantum mechanics and identify conditions that favor significant tunneling effects.

II. THEORY

The theoretical framework of this project is based on the time-dependent Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x)\psi$$

Where $V(x)$ represents the potential barrier defined by a step function at $x = 0.6$ with width d and height $V_0 = 9.80 \times 10^5$. The wave packet is initially modeled as a Gaussian centered at $x_0 = 0.3$ with a momentum represented by $k_0 = 700$ and a spatial width σ .

III. METHODOLOGY

This project employs the Crank-Nicolson numerical algorithm, a widely used method for solving partial differential equations like the time-dependent Schrödinger equation. The Crank-Nicolson method is chosen for its numerical stability and second-order accuracy, both crucial for accurately capturing the dynamics of quantum tunneling.

A. Crank-Nicolson Method

The Crank-Nicolson method involves a finite difference scheme that is implicit in time and centered in space. This method averages the time-forward and time-backward Euler methods, yielding a tridiagonal matrix equation at each time step, which balances stability with computational efficiency.

a. Setup and Implementation The one-dimensional spatial domain is discretized into a grid of 2500 points with a spatial step of $dx = 5.0 \times 10^{-4}$ and a temporal step of $dt = 5.0 \times 10^{-7}$. The Gaussian wave packet is initialized based on its position, width, and momentum parameters. A potential barrier is introduced at specified positions on the grid, influencing the wave packet as it evolves.

The wave function $\Psi(x, t)$ and the potential $V(x)$ are represented on this grid. The time evolution of $\Psi(x, t)$ is governed by the Time-Dependent Schrödinger Equation (TDSE) discretized using the Crank-Nicolson scheme. The TDSE involves the construction of matrices A and B , which are derived from the discretized Hamiltonian operator incorporating the kinetic and potential energy terms. Matrix A represents the system at the next time step, and matrix B incorporates the state at the current time step.

b. Matrix Construction The matrices A and B are constructed to evolve the wave function over time. Each diagonal and off-diagonal element of these matrices is calculated as follows:

$$\begin{aligned} A(k, k) &= 1 + 2i \cdot q + i \cdot \frac{dt}{2} \cdot V(k), \\ A(k, k-1) &= A(k, k+1) = -i \cdot q, \\ B(k, k) &= 1 - 2i \cdot q - i \cdot \frac{dt}{2} \cdot V(k), \\ B(k, k-1) &= B(k, k+1) = i \cdot q, \end{aligned}$$

where $q = \frac{dt}{4 \cdot dx^2}$, and $V(k)$ is the potential at the k -th grid point. This setup ensures that the Crank-Nicolson method accurately accounts for both the kinetic and potential energy terms in the Hamiltonian.

c. Boundary Conditions Boundary conditions are crucial for the stability and accuracy of the simulation. At the edges of the spatial domain, the wave function is constrained to prevent unphysical reflections or numerical instabilities. This is implemented by setting the first and last rows of matrices A and B to enforce

$$\Psi(0, t) = \Psi(K_{\text{space}}, t) = 0.$$

d. Numerical Solution To evolve the wave function $\Psi(x, t)$ according to the TDSE, $A \cdot \Psi(n+1) = B \cdot \Psi(n)$, the system is solved iteratively at each time step. The matrix A is tridiagonal, facilitating efficient numerical inversion using sparse matrix techniques. This efficiency is crucial for managing the computational load, given the iterative nature of the method over thousands of time steps.

e. Verification and Analysis The methodology includes steps for verifying the numerical solution, such as checking the conservation of total probability and tracking the wave packet's expected position and spread over time. These checks ensure that the simulation behaves physically and provides reliable insights into the quantum tunneling phenomenon.

IV. VERIFICATION OF NUMERICAL METHODS

The Crank-Nicolson method, used to solve the time-dependent Schrödinger equation for the quantum tunneling simulation, requires rigorous verification to ensure the accuracy and stability of the results. The following tests are performed with the potential $V = 0$ everywhere, enabling the wave packet to propagate freely, which simplifies the analysis of fundamental properties of the wave packet without the influence of external potentials.

A. Conservation of Total Probability

The total probability, represented by the integral of the probability density over all space, $\int |\Psi(x, t)|^2 dx$, is calculated at each time step. For a physically accurate simulation, this quantity should remain constant and equal to 1.0 throughout the simulation, confirming that the wave function is properly normalized and that the numerical method conserves probability. Figure 1 displays this behavior, as expected.

B. Expected Position

The expected position $\langle x \rangle$, defined as $\int x |\Psi(x, t)|^2 dx$, provides insights into the motion of the wave packet's center. As the wave packet propagates, $\langle x \rangle$ should linearly increase over time, corresponding to the wave packet moving with a velocity approximately equal to k_0 , the initial momentum of the wave packet divided by the mass of the particle. Figure 2 verifies that the particle behaves as expected. The parameters were set such that $k_0 = 575$, and this was cross-referenced by the slope of the position, which was calculated to be 565.5. Although it is not an exact match, it is sufficiently close enough to accurately observe quantum tunneling behavior.

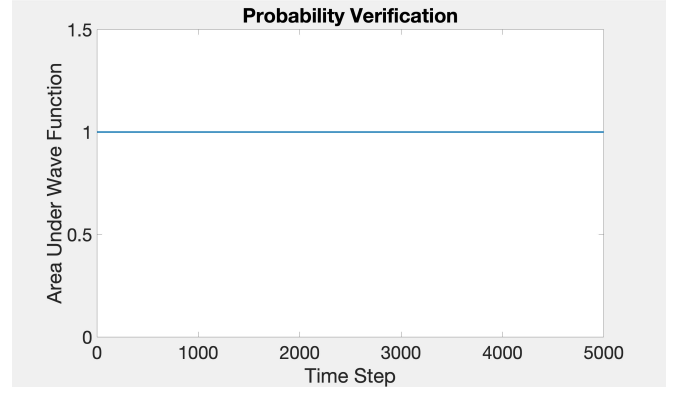


FIG. 1: Plot of Total Area under Wave Function vs. Time, showing that total probability equals unity throughout the entire simulation.

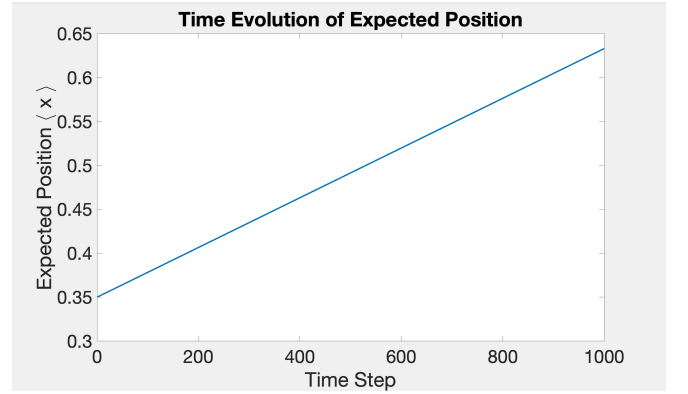


FIG. 2: Plot of Expected Position vs. Time of a free particle for $k_0 = 575$. Position varies linearly with time, as predicted.

C. Wave Packet Spread and Variance

To analyze the dispersion of the wave packet, the expected square position $\langle x^2 \rangle$ is calculated using $\int x^2 |\Psi(x, t)|^2 dx$. From this, the variance of the wave packet's position, $\sigma^2(t) = \langle x^2 \rangle - \langle x \rangle^2$, is derived. This variance, $\sigma(t)$, indicates how much the wave packet spreads as it propagates. According to quantum mechanics, narrower wave packets spread faster than wider ones due to their higher initial momentum uncertainty. Figure 3 depicts this expectation, where two particles with different initial spread both increase spread over time, with the narrower packet spreading faster than the wider one.

V. RESULTS

The first study is done with a single barrier, using the parameters $\sigma = \sqrt{2.5 \times 10^{-4}}$, $k_0 = 700$, $x_0 = 0.3$ and a barrier potential of $V_0 = 9.80 \times 10^5$. Figure 4 shows a snapshot of the wave packet at 3 different time steps.

We allow the barrier width to vary to see how it affects

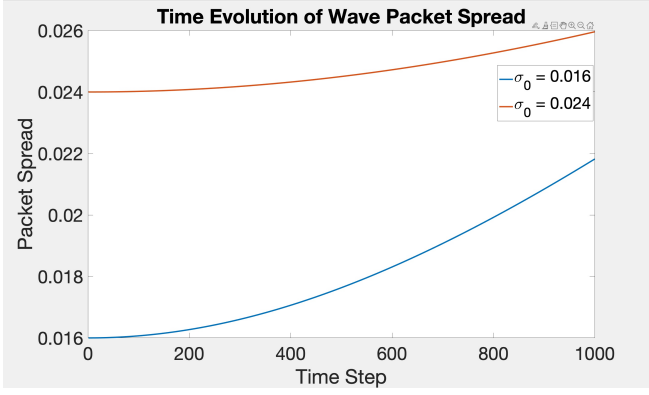


FIG. 3: Plot of wave packet spread over time of two wave packets with different initial spread.

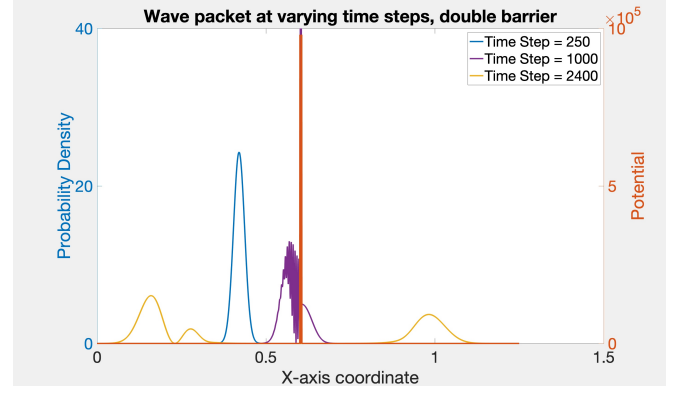


FIG. 6: caption.

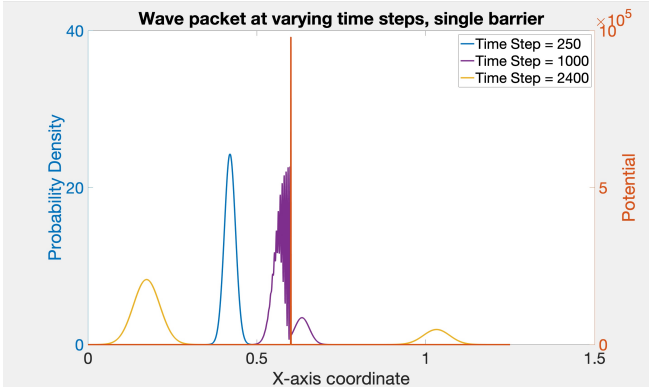


FIG. 4: caption.

the transmission coefficient. The results are shown in Figure 5.

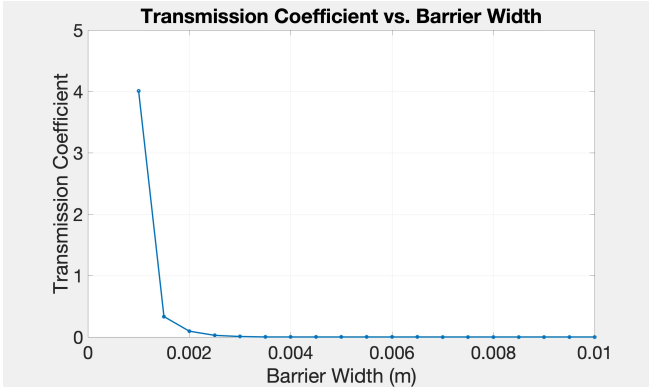


FIG. 5: caption.

The next study is done with a double barrier, using parameters $\sigma = \sqrt{0.004}$ and $x_0 = 0.3$. The barrier was 0.001 thick, spaced 0.004 apart, with the same potential as the prior study. Figure 6 shows a snapshot of the wave packet at 3 different time steps.

We vary the initial momentum to vary to see how it affects the transmission coefficient. The results are shown

in Figure 7.

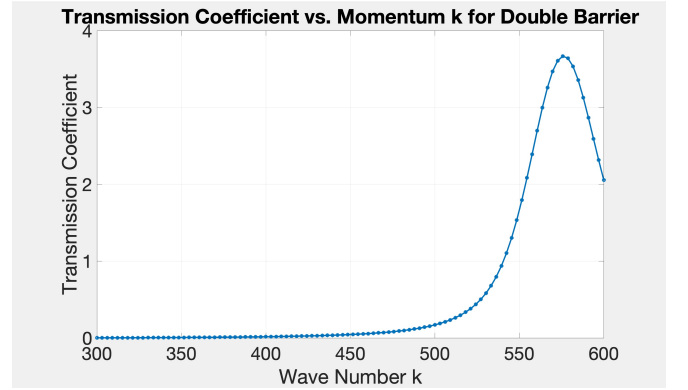


FIG. 7: caption.

VI. DISCUSSION

This study provides insightful results on quantum tunneling by analyzing the transmission coefficients as functions of barrier width and momentum. The graphs and corresponding analytical predictions help to understand the underlying quantum mechanics involved in tunneling through single and double barriers.

A. Transmission Coefficient vs. Barrier Width

The "Transmission Coefficient vs. Barrier Width" graph shows a sharp decrease in the transmission coefficient as the barrier width increases from zero, quickly approaching zero as the width exceeds a small threshold. This behavior is consistent with quantum mechanical predictions, where increasing the barrier width significantly reduces the probability of tunneling, due to the exponential decay of the wave function within the barrier. The initial high value reflects the near certainty

of transmission for very thin barriers, where the barrier presents almost no obstacle to the wave packet.

B. Transmission Coefficient vs. Momentum for Double Barrier

The "Transmission Coefficient vs. Momentum k for Double Barrier" graph exhibits a pronounced resonance peak, indicating conditions under which the transmission coefficient dramatically increases, despite the presence of barriers. This resonance is a phenomenon of quantum mechanics known as quantum resonant tunneling. It occurs when the wave packet's energy aligns with one of the quantized energy states that can form in the well between the two barriers. At these specific energies, the probability of tunneling increases significantly, leading to peaks in the transmission coefficient.

The observed resonance shows the wave-like nature of quantum particles, which can constructively interfere under the right conditions to enhance tunneling probabilities. This is particularly evident in the double barrier setup, where the spacing between barriers allows the formation of standing wave patterns that match the energy of the incoming wave packet.

C. Implications and Further Analysis

The results demonstrate how quantum tunneling deviates from classical expectations, where a particle would require enough kinetic energy to overcome a barrier. Instead, quantum mechanics allows for tunneling with probabilities influenced by the barrier's properties and the particle's wave characteristics.

Further analysis could involve studying the effects of varying other parameters such as barrier height or the initial position of the wave packet. Additionally, exploring different barrier shapes or more complex potential landscapes could provide deeper insights into the nuances of quantum tunneling and its applications in modern technologies, such as quantum computing and semiconductors.

These studies not only deepen our understanding of quantum mechanics but also pave the way for engineering novel quantum devices that leverage tunneling effects at the quantum scale..

VII. CONCLUSION

This study demonstrates quantum tunneling in a one-dimensional system, highlighting the deviations from classical predictions and providing insight into the wave packet's behavior under quantum mechanical rules.

[1] Author, *Title*, Journal, Volume, Page, Year.